Progress Check: PPE Safety Equipment Detection

P14 — 27 October 2026

Work completed and evaluation protocol

Both models use all 11 Construction-PPE classes and the frozen 1265/73/78 split. The training protocol is 50 epochs, batch 16, 640-pixel inputs, SGD lr0=0.01, momentum 0.937, weight decay 0.0005, three warm-up epochs and cosine decay; seeds are 42, 43 and 44. Validation inference uses fixed square letterboxing, class-aware NMS IoU 0.70 and a low confidence cutoff of 0.001 for AP. We sweep confidence 0.05 to 0.90 in steps of 0.05. Detections are matched one-to-one within each class at IoU $\geq$0.50 in descending IoU order. One threshold per model maximizes mean validation macro-F1 across seeds; ties favour the lower threshold. Classes without ground truth are excluded from macro averages and their false positives remain counted. No test predictions are used in this progress check. To avoid the observed MPS indexing fault, the unchanged NMS algorithm runs on CPU for both models; timing includes CPU/GPU transfers.

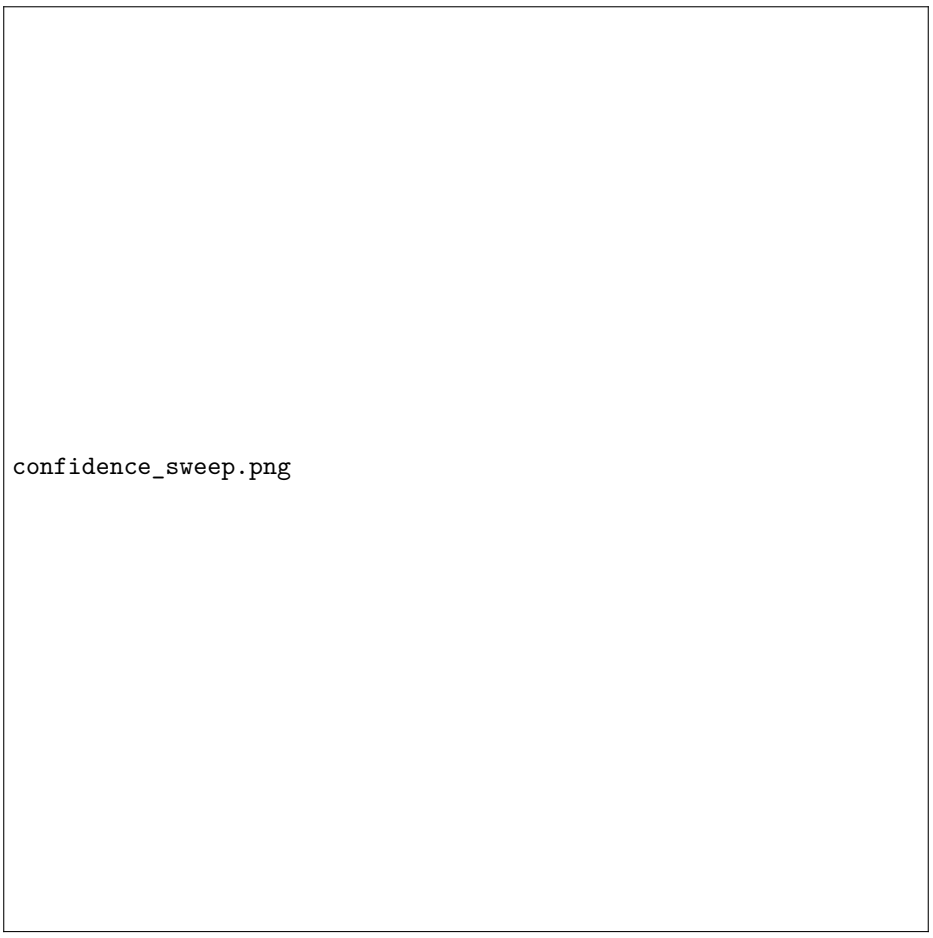
Preliminary validation comparison

Metric	YOLO11n	YOLO11s
mAP@0.50:0.95	0.1938 ± 0.0111	0.2043 ± 0.0080
mAP@0.50	0.4716 ± 0.0291	0.4966 ± 0.0119
Macro precision	0.5128 ± 0.0559	0.5767 ± 0.0156
Macro recall	0.4998 ± 0.0149	0.4840 ± 0.0199
Macro-F1	0.4962 ± 0.0253	0.5182 ± 0.0155
Macro-F1 at confidence 0.25	0.4882 ± 0.0265	0.5132 ± 0.0160

Confidence thresholds: YOLO11n 0.20, YOLO11s 0.30.

Mean ± sample SD across three seeds. Precision, recall and macro-F1 use the selected model threshold; AP uses ranked predictions.

In the common validation re-evaluation, the YOLO11s minus YOLO11n difference in mean mAP@0.50:0.95 is +0.0104. With only 73 validation images and three seeds, confidence in this preliminary difference remains limited. The validation set contains only eight goggles and four no_boots instances.



Batch-one median predict-call latency was 7.58 ms for YOLO11n and 10.03 ms for YOLO11s, on the same Apple M2 Max device. Each model used 20 warm-up calls and 100 measured calls, FP32 and confidence 0.25. Timing includes preprocessing, forward inference, NMS and predict-call overhead; disk IO and weight loading are excluded.

Relationship to the previous report

The submitted Stage 3 report recorded YOLO11s mAP@0.50:0.95 0.2062 +/- 0.0097 on Tesla T4 with Ultralytics 8.4.182. Those original weights and logs were not retained. We therefore trained a new three-seed YOLO11s reproduction run and retain its real outputs. The historical report remains unchanged; its values are not attached to the new checkpoints. This table re-evaluates all six available checkpoints in one environment. Unlike library-default rectangular training validation, this comparison explicitly uses fixed square letterboxing, so its AP can differ from the training summaries. The earlier F1 row was not an explicit macro-F1 evaluation; Stage 4 now averages class F1 scores directly. The reproduction trained 3 independent seed processes concurrently on one device; per-seed training wall times include contention. PyTorch reports some nondeterministic MPS kernels despite the deterministic flag; identical seeds do not guarantee bitwise-identical results. Because native MPS sparse indexing failed in initial attempts, the clean reproduction uses the original CPU detection criterion with differentiable device copies. Model forward/backward stay on GPU; loss values and gradients were checked against CPU. Interrupted attempts are excluded from this table.

Common evaluation environment: Python 3.14.7, PyTorch 2.14.1, Ultralytics 8.4.175; macOS-15.7.2-arm64-arm-64bit-Mach-O.

Remaining experiments

- Freeze validation-selected thresholds and evaluate all six checkpoints on the held-out test split.
- Report per-class AP and recall, and compare small/medium/large object recall with group counts.

- Compare clear/occluded recall and size distributions; disclose that current occlusion tags are a single assistant review.
- Explain at least three actual errors: a small-object miss, an occlusion-related miss, and a false positive or localization error.
- Complete the two-page final summary, reproducibility checks, and the five-minute demo with at most three slides.

Repository: `https://github.com/Ta6leTkaa/intro-to-CV-Project`